Conserv Genet (2009) 10:453–463
DOI 10.1007/s10592-008-9623-0

RESEARCH ARTICLE

Genetic structure of wild populations of the endangered Desert Pupfish complex (Cyprinodontidae: *Cyprinodon*)

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Received: 18 January 2008 / Accepted: 27 May 2008 / Published online: 5 June 2008
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Abstract Data from 10 microsatellite DNA loci were used to describe the genetic structure of the two extant species (*Cyprinodon macularius* and *C. eremus*) of the endangered Desert Pupfish complex of southwestern United States and northwestern Mexico. Variation at microsatellite loci was significantly correlated (Mantel test) with that of previous mtDNA results, both for the complex and for the relatively wide-ranging *C. macularius* alone. Both species showed unusually high levels of microsatellite diversity for non-marine fish ($H_e = 0.84$–0.93; $A_R = 11.9$–17.0). There was evidence ($R_{ST} > F_{ST}$) that the two extant populations of *C. eremus* have been isolated sufficiently long for mutation to contribute significantly to genetic divergence, whereas divergence among the nine assayed populations of *C. macularius* could be attributed to genetic drift alone. Correspondingly, 10% of the diversity in *C. eremus* was attributable to differences between the two populations, whereas, for *C. macularius*, only 2.7% was attributable to among-population variation. Within *C. macularius*, a small (0.8%), but statistically significant, portion was attributable to differences between populations in the Salton Sea area and those on the lower Colorado River delta. The two populations of *C. eremus* and five groups of populations of *C. macularius* are recommended as management units for conservation genetics management of the two species.

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Keywords *Cyprinodon macularius* · *Cyprinodon eremus* · Pupfish · Conservation genetics · Population structure

Introduction

Conservation of native, non-game fishes has been said to have begun with Carl Hubbs' campaign in the 1940 s to protect the Devil's Hole Pupfish, *Cyprinodon diabolis*, and its habitat (Deacon and Williams 1991). Since then, and despite extensive legislation and management action aimed at conservation, native fish in the American Southwest, including pupfishes (Minckley et al. 1991; Echelle et al. 2003), continue to decline as a result of anthropogenic habitat alteration and introductions of non-native fishes (Minckley and Deacon 1991; Contreras and Lozano 1994). One of the most dramatic declines has been the status of the Desert Pupfish complex (Cyprinodontidae), a once wide-ranging group that has disappeared from perhaps 95% of its historical range in the past several decades (Miller and Fuiman 1987; Hendrickson and Varela-Romero 1989). This complex has arguably received more attention from conservation managers than any other non-game species of fish (Minckley et al. 1991).

In this paper, we use variation at microsatellite loci to assess the genetic structure of the remaining wild populations of the Desert Pupfish complex. The complex comprises three species, the now extinct Monkey Springs Pupfish, *Cyprinodon arcuatus*, of southern Arizona (Minckley et al. 2002) and two extant species, the Desert Pupfish, *C. macularius*, which once ranged widely in the Colorado River system of southern Arizona and California and northern Sonora and Baja California del Norte, and *C. eremus*, which is known only from Quitobaquito Springs

in southern Arizona and a short reach of the Río Sonoyta in northern Sonora (Miller 1943; Hendrickson and Varela-Romero 1989; Echelle et al. 2000). Because of dramatic declines in distribution and abundance, the Desert Pupfish complex is listed (as *C. macularius*) as endangered by the United States (Federal Register 51(61):10842) and Mexico (31 July 2007; http://www.conabio.gob.mx/conocimiento/ise/fichasnom/).

The prognosis for the existing wild populations of the Desert Pupfish complex is poor (Dunham and Minckley 1998). Consequently, state and federal agencies have established more than 40 populations in quasi-natural refuges ranging in surface area from ~ 3 m^2 to ~ 6000 m^2 in facilities as diverse as a national fish hatchery, national wildlife refuges, public schools and universities, state parks, and private zoos and arboretums in California, Arizona, New Mexico, and Sonora (Koike 2007). Two important purposes of such stocks are to protect genetic resources against catastrophic loss of natural populations, and to provide stocks that might eventually be used for reintroductions into the historical ranges of the two species. To serve these purposes, captive populations should be managed to preserve, as closely as possible, the original genetic diversity of the species. Thus, there is a need for documentation of the pattern of diversity in wild populations.

Previous studies of genetic variation in natural populations of the Desert Pupfish complex consist of allozyme surveys of populations in the United States (Turner 1983; Dunham and Minckley 1998) and a mitochondrial DNA (mtDNA) survey throughout the range of the complex, including the known populations in Mexico (Echelle et al. 2000). In this paper, we provide a survey of microsatellite DNA variation in the samples assayed for mtDNA variation by Echelle et al. (2000). Our purposes are to (1) compare the pattern of genetic variation with results from previous studies of the complex and (2) use the results as a basis of recommendations for genetic management of the group.

Taxonomy and recent history of the complex

Miller and Fuiman (1987) described *C. m. eremus* as a subspecies endemic to Quitobaquito Springs, with the nearby Río Sonoyta population being morphologically intermediate between that subspecies and the remainder of *C. m. macularius*. Subsequently, on the basis of mtDNA variation and color pattern in breeding males, Echelle et al. (2000) elevated the subspecies names to species status, with *C. eremus* comprising populations in the Río Sonoyta basin, including Quitobaquito Springs, and *C. macularius* comprising all other populations of the complex. The phylogenetic position of the extinct *C. arcuatus* of the upper Santa Cruz River basin in southern Arizona has not been determined, but, based on geographic distribution and morphology, it is assumed to belong to the Desert Pupfish complex (Minckley et al. 2002).

Construction of upstream dams and water diversions on the Colorado and Gila rivers has eliminated most of the flow that once supported an expansive system of shallow-water pupfish habitats on river floodplains and the Colorado River delta at the head of the Gulf of California (Hendrickson and Varela-Romero 1989; Mueller and Marsh 2002). Elsewhere, including the Río Sonoyta in Sonora, and tributaries of the Gila River in Arizona, overgrazing by livestock, over-pumping of groundwater, and other anthropogenic factors caused extensive erosion, channel entrenchment, and drying of the shallow marshy habitats occupied by the species (Minckley 1973; Miller and Fuiman 1987). Such factors, together with the introduction of non-native fishes, led to the extinction of *C. arcuatus* 23 years prior to description of the species (Minckley et al. 2002).

Historical collection records of *C. macularius* are sparse, making it difficult to compare past and present distributions (Minckley 1973; Hendrickson and Varela-Romero 1989). Regardless, records starting in the 1940's (Miller 1943) indicate a declining trend in distribution and abundance, with *C. macularius* extirpated from as much as 95% of its historical range, primarily as a result of habitat loss and competition with non-native fishes (Moyle 1976; Hendrickson and Varela-Romero 1989; Martin and Saiki 2005).

The history of the Salton Sea includes an irregular cycle of inundation by overflow from the Colorado River, followed by gradual desiccation when the river flow returned to the Gulf of California. According to Carpelan (1961), the river overflowed into the Salton Sink seven times between 1849 and 1907. The present Salton Sea originated in 1905–1907 when, during a period of high flow, the Colorado River emptied into the Salton Sink, an event that would have inundated springs and marshes supporting *C. macularius* (Miller and Fuiman 1987). In the early 1960 s the pupfish was abundant in shoreline pools of the Salton Sea (Barlow 1961). Now, populations are sparse, but still occur along the shoreline of the Salton Sea and in San Felipe and Salt creeks, various irrigation canals and drains, and San Sebastian marsh at the southwestern corner of Salton Sea (Martin and Saiki 2005).

The largest populations of *C. macularius* now occur in springfed marshes and artificial pools and canals in the vicinity of the Cerro Prieto geothermal field, where salinities often are too high to support non-native fishes (Varela-Romero et al. 2002). This population is in the Rio Hardy watershed, which drains directly to the Gulf of California.

Historically, however, the low divide separating Cerro Prieto from the Salton Sink was occasionally breached during high flows in the Colorado River, when the river emptied into the Salton Sea via "Volcano Lake," a shallow wetland encompassing the Cerro Prieto area (MacDougal 1906; Sykes 1926).

The range of *C. eremus* once encompassed approximately 100 km of the Río Sonoyta, but now is restricted to about 1–2 km of the river and the nearby (2 km) Quitobaquito Springs. The Quitobaquito Springs population consists of several thousand adults in a spring-fed habitat in Organ Pipe Cactus National Monument (Douglas et al. 2001). The spring flows have been declining since the 1970 s, and in September and October of 2007, apparently because of a leak in the dam or elsewhere, the water depth in the receiving pond declined dramatically from 75 cm and stabilized at about 10 cm (T. Tibbetts, pers. comm.). Persistence of the Rio Sonoyta population is tenuous because the occupied stream-reach is subject to desiccation during drought (C. O. Minckley, pers. comm.). Miller and Fuiman (1987) concluded from fossil spring deposits that the two populations probably had been in permanent contact at "some time in the Holocene and certainly during Pleistocene and earlier times." They wrote that "field reconnaissance" contradicted Miller's original assumption that Quitobaquito Springs and the Río Sonoyta shared a direct water connection sometime within the past 100 yr.

Methods

DNA samples used in this study were those used by Echelle et al. (2000) from collections made in 1997 and 1998 at 11 sites ($N = 20$–25) representing all known wild populations of *C. macularius* and *C. eremus*. Localities were as follows (parentheses = locality numbers in Fig. 1): (1) county-line drain, 100 m from shore of Salton Sea at boundary between Riverside and Imperial Counties, California (33° 29′ 56″ N, 106° 04′ 45″ W); (2) San Felipe Creek, Imperial County, California (33° 06′ 01″ N, 115° 55′ 57″ W); (3) Shoreline pool of Salton Sea near Trifolium 20A drain, Imperial County, California (33° 29′56″ N, 106° 04′ 45″W); (4) A slough near Cerro Prieto and 0.8 km N of a geothermal power plant, Baja California (32° 25′ 26″ N, 115° 15′ 44″ W); (5) Pozo del Tules, at east edge of Laguna Salada, 20.6 km S of Highway 2, Baja California (32° 26′ 27″ N, 115° 36′ 15″ W); (6) Santa Clara Slough at terminus of Canal Sanchez Taboada, Sonora, Mexico (32° 03′ 28″ N, 114° 53′ 50″ W); (7) canal at Flor del Desierto, Sonora (32° 01′ 56″ N, 114° 51′ 36″ W); (8) spring at El Doctor, Sonora (31° 56' 12" N, 114° 43' 56" W); (9) spring about 150 m NW of locality 8 (31° 56′ 47″ N, 114° 44″ 56″ W); (10) Quitobaquito Springs/Pond, Organ Pipe Cactus National Monument, Pima County, Arizona (31° 56′ 24″ N, 113° 00′ 44″ W); (11) Río Sonoyta 11.5 km W, 1.5 km S of Sonoyta, Sonora, Mexico (31° 54′ 05″ N, 112° 57′ 41″ W).

We assayed 10 microsatellite loci, nine (listed below) from Burg et al. (2002), and one (WSP–02) from Stockwell et al. (1998). Microsatellites were amplified by polymerase chain reaction (PCR) in 15-µl reactions comprising 9.0 µl True Allele PCR Premix (Applied Biosystems Inc.), 3.8 µl ddH_2O, 1.0 µl of each primer (5.0 µM), and 1.2 µl template DNA. The reverse GATA73 primer and the forward primer for the remaining loci were end-labeled with a fluorescent dye as follows: GATA73, HEX; GATA2, 6-FAM; GATA5, HEX; GATA9, NED; GATA10, 6-FAM; GATA26, HEX; GATA39, NED; CmD1, NED; CmD16, 6-FAM; WSP-02. Amplifications were done in either an MJ Research PTC 100 Thermal Cycler or a Perkin Elmer 9600 Thermal Cycler. Annealing temperatures are given in Table 1. Reaction conditions for the PTC 100 consisted of one cycle at 95°C for 12 min; five cycles of 94°C for 30 s, T_{A1} for 30 s, 72°C for 30 s; 32 cycles of 94°C for 30 s, T_{A2} for 30 s, 72°C for 20 s; and one cycle at 72°C for 2 min. Reaction conditions for the PE-9600 consisted of one cycle at 95°C for 12 min; five cycles of 94°C for 30 s, T_{A1} for 30 s, 72°C for 30 s; 35 cycles of 94°C for 45 s, T_{A2} for 45 s, 72°C for 1 min; and one cycle at 72°C for 2 min. After amplification, we mixed 1 µl from each reaction with 2 µl formamide and 0.5 µl of ROX-labeled size standard

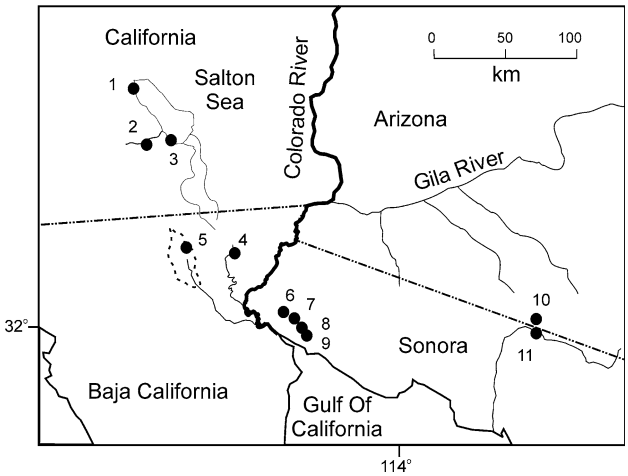

Fig. 1 Collection localities for wild populations of the Desert Pupfish complex. Localities 1–9 represent the remaining populations of *C. macularius*, which historically occurred in the Gila River of Arizona and the lower Colorado River of California and Arizona. Localities 10 and 11 represent the two populations of *C. eremus*. Locality abbreviations used in the text are as follows: 1 = County Line Drain (CLD), 2 = San Felipe Creek (SFC), 3 = shoreline pool of Salton Sea (SPSS), 4 = Cerro Prieto (CP), 5 = Pozo del Tules in Laguna Salada (LS), 6 = Canal Sanchez Taboada (CST), 7 = Flor del Desierto (FDD), 8 = El Doctor 1 (ED1), 9 = El Doctor 2 (ED2), 10 = Quitobaquito Springs (QS), 11 = Río Sonoyta (RS)

Table 1 Characteristics of primers used in this study

Locus	T_{A1}/T_{A2}[a]	Marker size range
GATA2	45/48	207–351
GATA5	50/53	181–269
GATA9	50/53	237–357
GATA10	50/53	212–332
GATA26	50/53	200–280
GATA39	50/53	228–348
GATA73	40/43	276–324
CmD1	58	218–370
CmD16	58	344–424
WSP-02	55	204–294

[a] T_{A1} and T_{A2} are the first and second annealing temperatures (see text)

WSP-02 has dinucleotide repeats, the GATA and CmD loci have tetranucleotide repeats

GATA and CmD primers are from Burg et al. (2002); WSP-02 is from Stockwell et al. (1998)

and resolved the PCR product using an automated sequencer (Applied Biosystems Inc., ABI 377 or Prism 3130) with Genescan (version 3.1); allele sizes were determined with the aid of Genotyper (version 2.5). In addition to using the size standard, at least one previously genotyped sample was included with nearly every set of samples to identify machine drift in output of allele sizes and to adjust for differences between machines.

MICRO-CHECKER (version 2.2.0.3; van Oosterhout et al. 2004) was used to identify genotyping errors due to null alleles, short allele dominance, and the scoring of stutter peaks. Brookfield's (1996) Method I was used to estimate null allele frequencies. We used GENEPOP (version 3.4; http://genepop.curtin.edu.au) to test for linkage disequilibrium and deviation from Hardy-Weinberg Equilibrium (Markov chain parameters = 1000 de-memorization, 100 batches, and 1000 iterations). We used GENEPOP to compute observed heterozygosity (H_O), expected heterozygosity (H_E), GDA (Lewis and Zaykin 2001) to obtain number of alleles (A) and allele richness (A_R), and CONVERT (Glaubitz 2004) to obtain number of private alleles (A_P). The sequential Bonferroni correction (Rice 1989) was used to reduce Type I errors for multiple tests applied to the same hypothesis.

We used ARLEQUIN (version 2.001; Schneider et al. 2000) for analysis of molecular variance (AMOVA; Excoffier et al. 1992) on both the microsatellite data and the mtDNA sequence data from Echelle et al. (2000). We used GDA and RstCALC (Goodman 1997) to compute, respectively, pairwise F_{ST} and R_{ST} values and, based on bootstrapping across loci (10,000 pseudo-replicates), their 95% confidence intervals (CI) among populations. We used SPAGeDi (Hardy and Vekemans 2002) for an allele-size permutation test (5000 iterations) to assess whether differences in microsatellite allele size (mutation) contributed to genetic divergence ($R_{ST} > F_{ST}$) or whether divergence can be attributed to genetic drift alone. When $R_{ST} > F_{ST}$, the appropriate measure of divergence is R_{ST}; it is F_{ST} when $R_{ST} \leq F_{ST}$ (Hardy and Vekemans 2002).

We used GENEPOP to perform a Mantel test of correlation between mtDNA and microsatellite divergence among populations (Echelle et al. 2000). Pairwise Φ_{ST} (mtDNA) and R_{ST} (microsatellites) values were used in the test across both species and the test for *C. eremus*. For *C. macularius*, we used F_{ST} for both mtDNA and microsatellites. To test for isolation-by-distance in *C. macularius* we used the ISOLDE program in GENEPOP (10,000 permutations) and tested for association between Slatkin's linearized F_{ST} estimates ($F_{ST}/(1 - F_{ST})$) and the natural logarithm of geographic distances (measured with Google Earth). Geographic distances were the shortest water-distances for the populations in the Salton Sea area and the populations on the Colorado River delta; distances between Salton Sea populations and those on the Colorado River delta included the straight-line distance between the southern-most Salton Sea locality (locality 3) and Cerro Prieto (locality 4) plus the water distances (around the Sea of Cortez) from there to the remaining sites on the delta.

The neighbor-joining algorithm in MEGA 3.1 (Kumar et al. 2004) was used to summarize pairwise genetic divergences based on microsatellites (5 shared loci; R_{ST}) and mtDNA (Φ_{ST}) across all populations of the two species. We used multidimensional scaling (with Guttman's stress) in SYSTAT (version 10, SPSS Inc.) to visualize among-population divergences (F_{ST}) for the seven loci assayed in *C. macularius*.

Results

One locus (GATA73) amplified in all except two specimens of *C. eremus* (both in Quitobaquito Springs), but did not amplify in *C. macularius*. Similar results were obtained in a subsequent analysis of captive populations (H. Koike, pers. comm.). Therefore GATA73 amplification appears effectively diagnostic of the two species. This locus was dropped from the analysis of *C. eremus* because of consistent evidence of null alleles at frequencies >0.15.

Nine loci were retained in the analysis (GATA2, GATA5, GATA9, GATA10, GATA26, GATA39, CmD1, CmD16, and WSP-02). Two, GATA10 and GATA26, were not used for *C. macularius* because of consistent null-allele frequencies ≥10%, and two others, GATA2 and GATA5, were not used for *C. eremus* because of, respectively, inconsistent scoring and large heterozygote deficiencies. Removing two loci for each species left seven loci for

intraspecific analyses and five for analyses that included both species.

There was no evidence of linkage disequilibrium (without Bonferroni correction). With the Bonferroni correction, there was no evidence of HWE deviation, except for GATA5 at locality 2 (heterozygote deficiency, $P < 0.001$). Without the correction, there were six deviations ($P = 0.012$ to 0.048) in *C. macularius* (all heterozygote deficiencies). There was no statistically significant frequency of null alleles. Both species showed consistently high levels of genetic diversity (Table 2).

Based on the allele-size permutation test, $R_{ST} > F_{ST}$ when all populations of both species were included (5 loci; $P < 0.0001$) and for *C. eremus* alone (7 loci; $P = 0.02$), but not for *C. macularius* (7 loci; $P = 0.14$). Correspondingly, for *C. eremus* and the majority (11 of 19) of the interspecific values there was no overlap in 95% confidence intervals for F_{ST} and R_{ST} values in individual pairwise comparisons, whereas, within *C. macularius*, all F_{ST}-R_{ST} pairs showed overlapping 95% confidence intervals for the two indexes (Table 3). To explore the generality of these results across loci, we computed locus-by-locus R_{ST} and F_{ST} values for each of the two species and for the dataset comprising both species. Mean R_{ST} was higher for the seven loci surveyed in *C. eremus* (0.076; range = 0.000–0.245) than it was for the seven surveyed in *C. macularius* (0.029; 0.001–0.066), but the difference was not statistically significant (*t*-test; $P = 0.27$). Mean F_{ST} was 0.028 for both species (ranges: *C. eremus* 0.000–0.050; *C. macularius* 0.015- 0.054), and was not significantly different from R_{ST} for either species ($P \geq 0.22$). For the 5-locus dataset across both species, mean R_{ST} per locus (0.131; 0.012–0.348) was greater than F_{ST} (0.051; 0.022–0.154), but the difference was not statistically significant ($P = 0.14$).

Inspection of results for individual loci shows that the indication from the allele-size permutation test that $R_{ST} > F_{ST}$ across all samples of both species is a result of size variation at two of the five loci: GATA39 ($R_{ST} = 0.200$, $F_{ST} = 0.024$) and WSP02 ($R_{ST} = 0.348$, $F_{ST} = 0.154$). The similar indication within *C. eremus* is also a result of two loci: GATA39 ($R_{ST} = 0.178$, $F_{ST} = 0.048$) and GATA10 ($R_{ST} = 0.245$, $F_{ST} = 0.025$). For all other loci in those two sets of analyses and in the analyses for *C. macularius* alone, both R_{ST} and F_{ST} were relatively small (range = 0.000–0.053).

All interspecific, pairwise estimates of F_{ST} and R_{ST} were significant, as were those between the two populations of *C. eremus* (Table 3). Within *C. macularius*, none of the R_{ST} values were significant, but, based on F_{ST}, the three populations from the Salton Sea were significantly divergent from each other and from most populations on the lower Colorado River delta. The latter group showed few significant differences except for comparisons involving the Laguna Salada population (locality 5) and El Doctor 2 (locality 9).

Neighbor-joining trees grouped *C. macularius* and *C. eremus* separately on the basis of both mtDNA and microsatellite DNA (Fig. 2). Most alleles at the five microsatellite loci analyzed in both species were present in both species, as indicated by numbers of private alleles (Table 2). The locus most nearly diagnostic of the two species was WSP-02. Allele 204 was absent in 7 of the 9 populations of *C. macularius* and occurred only as single heterozyotes in the other two populations (allele frequency = 0.025). The frequency of the allele was 0.72 and 0.77 in the two populations of *C. eremus* and all individuals were either homozygous or heterozygous for it.

For the seven loci assayed in *C. macularius*, multiple dimensional scaling separated the three populations in the

Table 2 Population genetic statistics averaged over 7 microsatellite loci in *Cyprinodon macularius* and a separate suite of 7 loci in *Cyprinodon eremus*

	C. macularius									*C. eremus*	
	Salton Sea			Lower Colorado River Delta						Río Sonoyta Basin	
	1	2	3	4	5	6	7	8	9	10	11
N	19–20	18–19	24–25	19–20	20	20	19–20	20	20	19–23	20
H_o	0.878	0.882	0.897	0.883	0.879	0.871	0.892	0.950	0.843	0.830	0.850
H_e	0.870	0.922	0.914	0.914	0.892	0.900	0.926	0.923	0.925	0.840	0.842
A	13.3	15.6	17.1	14.6	13.0	15.6	16.9	16.6	16.0	14.3	11.9
A_R	11.5	13.0	14.1	14.0	11.7	13.4	14.8	14.2	13.5	10.6	8.8
A_P	4/4	8/5	8/5	4/1	0/0	5/4	6/4	2/1	4/1	46/4	29/1

Locality numbers are as in Fig. 1. N = sample size, H_o = mean observed heterozygosity, H_e = mean expected heterozygosity, A = mean number of alleles, A_R = mean allele richness. A_P = number of private alleles in each species considered separately and (after the slash) when all populations of both species are considered together for the five shared loci

Table 3 Pairwise estimates (with 95% CI) of R_{ST} (above diagonal) and F_{ST} (below diagonal) among 11 wild populations (numbered as in Fig. 1)

	C. macularius									*C. eremus*	
	1	2	3	4	5	6	7	8	9	10	11
1		0.010	0.032	0.000	0.024	0.020	0.008	−0.003	0.029	**0.265**	**0.329**
		−0.007–0.010	0.008–0.115	−0.011–0.090	−0.003–0.118	0.000– 0.101	−0.011–0.090	−0.013–0.081	0.005–0.126	0.213–0.367	0.255–0.436
2	**0.033**		0.036	0.023	0.048	0.045	0.029	0.043	0.091	**0.350**	**0.393**
	0.024–0.054		−0.000–0.138	0.002–0.126	0.013–0.154	0.008–0.150	0.002–0.127	0.005–0.154	0.052–0.191	*0.285–0.450*	*0.312–0.501*
3	**0.029**	**0.007**		0.002	0.043	0.037	0.005	0.012	0.063	**0.324**	**0.391**
	0.021–0.037	0.001–0.021		−0.008–0.071	0.016–0.132	0.013–0.122	−0.008–0.073	−0.006–0.089	0.035–0.143	*0.265–0.413*	*0.314–0.487*
4	**0.020**	**0.016**	**0.017**		−0.001	0.041	−0.001	−0.003	0.023	**0.304**	**0.357**
	0.013–0.024	0.001–0.032	0.006–0.027		–0.016–0.076	0.019–0.130	−0.012–0.091	−0.007–0.076	0.000–0.121	0.233–0.407	*0.272–0.407*
5	**0.050**	**0.039**	**0.037**	**0.028**		0.056	0.030	0.010	0.042	**0.338**	**0.367**
	0.043–0.058	0.036–0.054	0.028–0.044	0.018–0.035		0.025–0.150	0.002–0.120	−0.007–0.093	0.013–0.128	*0.264–0.447*	*0.278–0.447*
6	**0.040**	**0.030**	**0.030**	**0.017**	**0.029**		0.047	0.003	0.066	**0.216**	**0.257**
	0.031–0.050	0.015–0.056	0.016–0.044	0.005–0.028	0.021–0.036		0.012–0.148	–0.008–0.085	0.035–0.154	0.174–0.300	0.202–0.362
7	**0.030**	**0.019**	0.015	0.008	**0.022**	0.015		0.005	0.021	**0.277**	**0.359**
	0.022–0.039	0.009–0.038	0.004–0.023	0.000–0.014	0.012–0.029	0.006–0.026		−0.007–0.081	0.003–0.103	0.208–0.387	*0.270–0.478*
8	**0.030**	0.007	0.013	0.008	0.016	0.014	0.004		0.027	**0.255**	**0.317**
	0.015–0.040	0.003–0.026	0.005–0.021	0.000–0.016	0.004–0.027	0.008–0.020	0.000–0.008		–0.000–0.120	0.188–0.358	*0.242–0.426*
9	**0.053**	**0.021**	**0.025**	**0.019**	**0.028**	**0.027**	0.009	0.008		**0.144**	**0.200**
	0.029–0.078	0.009–0.050	0.016–0.035	0.007–0.030	0.015–0.038	0.011–0.044	−0.003–0.022	−0.002–0.017		0.090–0.247	0.123–0.315
10	**0.128**	**0.119**	**0.118**	**0.097**	**0.108**	**0.103**	**0.093**	**0.099**	**0.104**		**0.100**
	0.039–0.290	*0.037–0.271*	*0.022–0.254*	0.014–0.249	*0.021–0.247*	0.018–0.248	0.009–0.235	0.018–0.245	0.018–0.255		*0.053–0.200*
11	**0.118**	**0.103**	**0.094**	**0.094**	**0.109**	**0.103**	**0.093**	**0.101**	**0.103**	**0.025**	
	0.042–0.245	*0.035–0.219*	*0.027–0.202*	*0.021–0.215*	*0.045–0.217*	0.031–0.219	*0.023–0.206*	*0.031–0.216*	0.030–0.228	*0.012–0.039*	

Within species values are based on 7 loci (different suite for each species), while among species values are based on 5 loci in common between the two species. Population numbers are as in Fig. 1. Bold values indicate statistical significance with the sequential Bonferroni correction (55 tests each half-matrix; <0.001 = initial P for significance at table-wide $\alpha = 0.05$). Italicized numbers indicate non-overlap in CI between R_{ST} and F_{ST} for individual comparisons

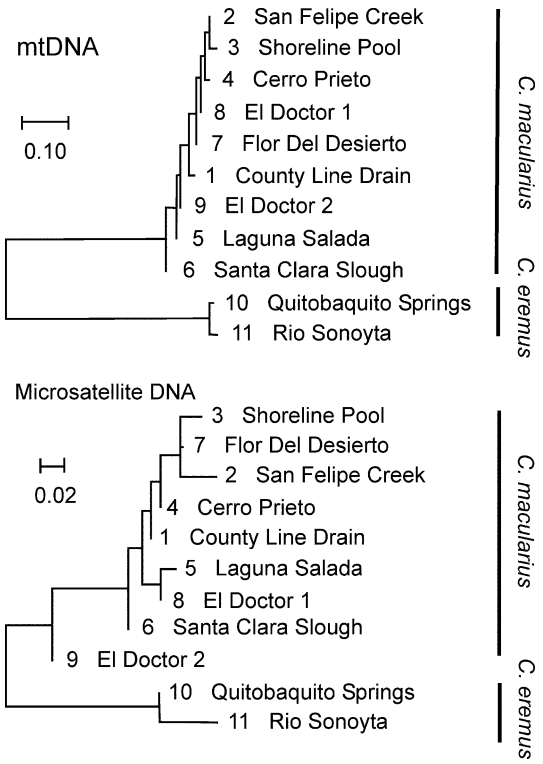

Fig. 2 Neighbor-joining trees based on pairwise Φ_{ST} (top) and R_{ST} (bottom) values. Numbers on terminal nodes are from Fig. 1. The microsatellite tree is based on the five loci in common between the two species

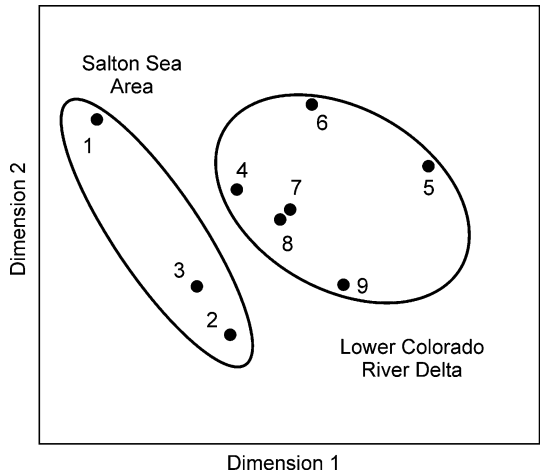

Fig. 3 Multidimensional scaling of pairwise F_{ST} values from microsatellite DNA variation in *C. macularius*. Based on the seven loci scored for this species. Population numbers are from Fig. 1. Percent of original variance explained = 96.6%. Guttman's stress, an inverse measure of correspondence between pairwise distances among plotted points in the figure and the original F_{ST} values = 0.11

Salton Sea area from the six on the lower Colorado River Delta, although the geographically intermediate population at locality 4 could have been placed in either group (Fig. 3). The Mantel test for isolation-by-distance in *C. macularius* was significant for microsatellites ($P = 0.0001$), but not for mtDNA ($P = 0.26$). Nonetheless, a tendency for parallel geographic variation in the two sets of data is indicated by the Mantel test of correlation between mtDNA and microsatellite genetic distances, which was significant, both across all populations of both species (5 microsatellite loci, $P = 0.003$) and for *C. macularius* alone (7 loci, $P = 0.048$).

AMOVA attributed 23% of the microsatellite diversity (R_{ST}) to differences between *C. macularius* and *C. eremus*, 2% to differences among populations within species, and 75% to variation within populations (Table 3). For *C. macularius* alone, a small (0.8%), but statistically significant portion of diversity (F_{ST}) was attributed to variation among groups (Salton Sea area vs lower Colorado River delta), 1.9% was attributed to variation among populations within groups, and 97.3% to variation within populations. For *C. eremus*, 10.0% and 90.0% of the diversity (R_{ST}) was attributed to, respectively, differences between the two populations and variation within populations. The AMOVA results for mtDNA generally corresponded closely with those for microsatellites (Table 3). The major discrepancy was with *C. eremus*, where, for mtDNA, variation reflecting divergence between the two populations was a statistically non-significant 2.9% ($P = 0.19$), whereas for microsatellites it accounted for 10% of the variation ($P < 0.00001$) (Table 4).

Discussion

Taxonomy

Species-level distinction of *C. macularius* and *C. eremus* is supported by reciprocal monophyly for mtDNA lineages (Echelle et al. 2000), subtle morphometic differences (Miller and Fuiman 1987), and a difference in intensity of yellow-orange pigmentation on the caudal fin and peduncle in breeding males (Echelle et al. 2000). Further, genotypes at two of the 10 loci assayed in this study allow effectively 100% separation of the two species. The two species showed an apparently fixed difference in presence/absence of a null allele for GATA73 genotypes, and, they were diagnostically different in WSP-02 genotype except for two individuals of *C. macularius*. A previous allozyme analysis found no diagnostic loci, but, based on allele frequencies, *C. eremus* from Quitobaquito Springs clustered outside of a group of five populations of *C. macularius* (Turner 1983). The indication that microsatellite allele-size contributes to divergence of the two species ($R_{ST} > F_{ST}$) demonstrates that divergence is not reflective of genetic drift alone and that the two species have been isolated sufficiently long that accumulation of new mutations has played a role

Table 4 Analyses of molecular variance in wild populations of the Desert Pupfish complex. Based on allele size diversity (R_{ST}) for microsatellites and sequence diversity (Φ_{ST}) for mtDNA. 1, 2, and 3 asterisks signify significance at, respectively, $P < 0.05$, $P < 0.005$, and $P < 0.00001$

Groupings	Percentage of total diversity attributable to differences:		
	Among groups	Among populations within groups	Within populations
C. macularius and *C. eremus*			
Microsatellites (5 loci), F_{ST}	8.0***	1.9***	90.0
Microsatellites (5 loci), R_{ST}	22.6***	2.0**	75.3
mtDNA, Φ_{ST}	86.1***	0.6*	13.3
C. macularius sites 1–3 & 4–9			
Microsatellites (7 loci), F_{ST}	0.8***	1.9***	97.3
Microsatellites (7 loci), R_{ST}	0.7	2.9***	96.4
mtDNA, Φ_{ST}	3.7*	2.1	94.2
C. eremus sites 10 & 11			
Microsatellites (7 loci), F_{ST}	2.5***	–	97.5
Microsatellites (7 loci), R_{ST}	10.0***	–	90.0
mtDNA, Φ_{ST}	2.9[a]	–	97.1

[a] $P = 0.19$

(Hardy et al. 2003). This reflects allele-size variation at particular loci, rather than being generally applicable across loci, presumably as a result of differences in locus-specific features such as mutation rate and degree of conformance to the stepwise mutation model. Molecular estimates of time of divergence indicate that the mtDNA lineages of the two species have been diverging for about 0.9 million years (Echelle 2008).

Overall genetic diversity

Both species of the Desert Pupfish complex show notably high levels of microsatellite diversity. Number of alleles per locus/population was 15.4 for *C. macularius* and 15.1 for *C. eremus*, and heterozygosity (H_e) per locus/population was, respectively, 0.91 and 0.84. These values are markedly higher than the averages (7.5 alleles/locus, $H_e = 0.46$) reported in a review of freshwater fishes (DeWoody and Avise 2000). Although comparisons involving different suites of loci must be tentative, the estimates for the Desert Pupfish complex are higher or similar to (H_e in *C. eremus*) those reported for the four most diverse populations of *C. nevadensis* (11.6 alleles/locus, $H_e = 0.83$), a species characterized as having abundant variation in both microsatellites and mtDNA (Duvernell and Turner 1998; Martin and Wilcox 2004). The estimates are markedly higher than reported for native populations of *C. tularosa* (4.0 alleles/locus; $H_e = 0.48$; Stockwell et al. 1998), a species having low allozyme variation in comparison with other pupfishes (Echelle 1991; Stockwell et al. 1998).

Intraspecific divergence

As suggested by Miller and Fuiman (1987), the Quitobaquito Springs and Río Sonoyta populations of *C. eremus* appear to have a long history of isolation, despite being separated by only about 2 km. This is consistent with (1) morphometric divergence of the two populations (Miller and Fuiman 1987), (2) the rather large proportion (10%) of microsatellite diversity attributable to differences between the two populations, and (3) the indication that, unlike *C. macularius*, mutation subsequent to isolation contributes significantly to the inter-population microsatellite divergence. In contrast, however, a statistically non-significant 2.9% of the mtDNA diversity in *C. eremus* reflected differences between the two populations. Failure to detect significant mtDNA divergence despite divergence in microsatellites might reflect, in part, the smaller sample of mtDNA markers (n = number of fish) compared with microsatellites (n = 2 X number of fish).

The genetic structure of *C. macularius* shows rather weak, albeit statistically significant, geographic structure. About 1% of the microsatellite diversity and 4% of the mtDNA diversity is attributable to differences between populations in the Salton Sea area and those from the lower Colorado River Delta, with 97% and 94% of, respectively, microsatellites and mtDNA diversity occurring within the average local population. This probably reflects gene flow across the Colorado River delta in historical times prior to the presently depleted river flow and the construction of levees that control the course of the river in Mexico.

The test for isolation-by-distance in *C. macularius* was statistically significant for microsatellites, but not for mtDNA, possibly reflecting the above-mentioned difference in sample size for the two types of markers. Presently, most of the populations are isolated by uninhabitable stretches of dry desert, and the Salton Sea populations have been isolated from those on the Colorado River delta for more than a century. But, historically, there was much greater connectedness among populations, including more frequent connections between the lower delta and the Salton Sea (Sykes 1926; Carpelan 1961). In those times, distance, and not geographical isolation per se, could have played a role in restricting gene flow among populations.

The population in Laguna Salada (Locality 5, Fig. 1) is perhaps the most isolated population of *C. macularius*, and, yet, unlike all other populations of the species it had no private alleles. Follett (1960) and Miller and Fuiman (1987) cited information from Carl Hubbs in reporting that the species had been extirpated from the basin by the 1960 s or 1970 s. In 1983–84, the species would have gained access to Laguna Salada when flows from the Colorado River temporarily filled the basin, the first such connection in more than 20 years, and by 1987 the resulting lake was reduced in size and desiccating (Hendrickson and Varela-Romero 1989). When the collection for our study was made in 1998, the lake was dry except for a small, spring-fed area (Pozo del Tules) on the northeast end of the basin that has persisted to the present and still supports a pupfish population (C. O. Minckley, pers. observ.). The lack of private alleles in our collection of this population might reflect recent colonization from elsewhere on the delta and genetic drift; microsatellite diversity, although high, was the lowest observed for the species.

Conservation

The remnant wild populations of both *C. macularius* and *C. eremus* retained large amounts of genetic variation at the time of our collections. This diversity in the face of severely reduced inter-population connectedness and numbers and sizes of populations probably is explained by life history traits. Most pupfishes are hardy, small-bodied omnivores capable of maintaining extremely dense populations, particularly in harsh habitats where other species are absent or uncommon (Echelle et al. 1972; Naiman 1976). Consequently, recent surveys found the densest populations of *C. macularius* in highly saline waters where they are least exposed to non-native fishes (Hendrickson and Varela-Romero 1989; Varela-Romero et al. 2002). Dunham and Minckley (1998, p. 13) commented that, "Fortunately, most non-native species… cannot survive the severe conditions pupfish accommodate readily." They suggested that this aspect of pupfish biology should be incorporated into conservation plans for *C. macularius*.

Results from this study, together with those from previous work (Echelle et al. 2000) support management of the Desert Pupfish complex as two separate species. Within species, there is no genetic evidence of separate evolutionarily significant units (ESUs), neither as defined by Waples (1991) on the basis of reproductive isolation and ecological or adaptive uniqueness, nor as defined by Moritz (1994) on the basis of reciprocal mtDNA monophyly.

Conservation biologists often use measures of genetic divergence such as F_{ST} to identify management units (MUs), which can be defined as demographically independent (isolated or relatively so) populations or groups of populations important for the long-term persistence of the species. This effort is confounded by two factors when based on highly variable loci like microsatellites (Hedrick 1999): (1) Because of high statistical power, detected divergence might not be biologically significant, and (2) on the other hand, when heterozygosity is high, as in the Desert Pupfish complex ($\sim$0.90), the among-population component of diversity will be small, even when populations share no alleles and clearly have experienced restricted gene flow for sufficient time to allow adaptive divergence. In addition, because of reduced genetic drift in large populations, they can be demographically independent while showing little divergence for neutral markers (Allendorf and Luikart 2007). Given these problems, we identify an MU for the Desert Pupfish complex as a population or network of populations that occupies a critical portion of the geographic range. We recommend recognition of two MUs in *C. eremus* (Quitobaquito Springs and Río Sonoyta) and five in *C. macularius*, three in the lower Colorado River delta (Laguna Salada, Cerro Prieto, and the Santa Clara Slough/El Doctor area) and two in the Salton Sea region (the San Felipe Creek/San Sebastian Marsh system and the remaining Salton Sea system). In our judgment, the loss of any one of these MUs would be a significant step toward extinction of the species in the wild.

There is some danger of misinterpreting the implications of MU recognition. Recognition of a set of populations (for example, shoreline pool and canal populations connected to the Salton Sea) as an MU does not mean that all except one of those populations are expendable or that we should preserve the most distinctive member of the MU. Instead, conservation management should aim to preserve the natural network of genetic connections between populations, thereby preserving the processes that maintain diversity and evolutionary potential (Crandall et al. 2000). For the Desert Pupfish complex, this network has been disrupted by human activities and local populations are now smaller and subject to extirpation or genetic bottlenecking, often with low potential for rescue by immigration from

elsewhere. With time, this will lead to reduced genetic diversity and evolutionary adaptability of the complex. Local populations should be monitored for abundance and genetic diversity and managers might eventually consider implementing a program of artificial gene flow among wild populations, in addition to the presently established refuge program for the complex.

Acknowledgements We thank G. Knowles, and B. Bagley for assisting with the Mexican field work, P. Barrett, A. F. Echelle, and S. Keeney for assistance in the field, and D. Duncan, and R. Clarkson for comments on an earlier draft of the paper. The work was supported through the Gila River Basin Native Fish Conservation Program of the U.S. Fish and Wildlife Service and the U.S. Bureau of Reclamation. The Oklahoma Cooperative Fish and Wildlife Research Unit provided administrative assistance.

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